Departamento de
Economía
Universidad de Concepción

INCAR^2
Interdisciplinary Center
for Aquaculture Research
- Applied Research

Are fishing decisions flexible?

Participation, target, and landing choices in the U.S. West Coast pelagic fishery

Felipe J. Quezada-Escalona

D. Tommasi · I.C. Kaplan · B. Muhling · S.M. Stohs

IIFET 2026 — Tórshavn — Spatial Trade-offs of EBM (Part 2), 20 Aug

NOAA CAFA (NA20OAR4310507) · ANID-Chile FONDECYT Iniciación N° 11250223 · INCAR^2 CIA250009.

Motivation

- Species distributions are **shifting** with changing ocean conditions — while fishable area shrinks (closures, offshore wind).
- For management, what matters is not the shock but how fleets **redistribute effort** — follow, switch species, switch ports, exit.
- **But** location-choice models take **one fishery at a time**: the **species-targeting** margin drops out *by construction*.
- **Research question:** how do **availability, prices, costs, weather** shape the **daily** joint choice to **participate**, **target a species** and **land at a port**? *How much flexibility do vessels actually have?*

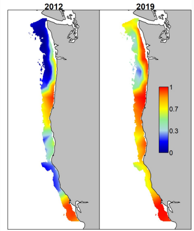
> **Spoiler:** where the boats end up is produced by **which species they can switch to**.

The contribution: an oceanographic SDM as the economic regressor

SDMs built by **oceanographers** for *ecological* purposes; their daily output is the central covariate in our **discrete-choice model of fishing behavior**.

Ocean state — ROMS: SST, chl-*a* → **GAM SDM** per species (Muhling et al. 2019, 2020)

→ **Daily** P(presence) on a spatial grid → Mean within the species **radius** of each port



Illustrative — Chasco et al. (2022)

$\mathrm{Avail}_{p,s,t}$: a measure of availability for *each* port × species × day $\Rightarrow$ **enters the fisher's utility directly**

Past catch vs. SDM

Past catch

- ✗ **Endogenous** — where vessels already *chose* to fish.
- ✗ **Selection bias** — only at fished sites (Chen et al., 2023).
- ✗ **Gaps** — undefined when nobody fishes.

SDM habitat index (ours)

- ✓ **Exogenous** — an environmental input.
- ✓ **No gaps** — a value for *every* port–day.
- ✓ **Counterfactual-ready**.

Assumption: habitat suitability tracks **potential catch rates** — if only partly, the availability effect is attenuated.

Setting — four fleet segments, four portfolios

U.S. West Coast Coastal Pelagic Species fishery: squid, sardine, anchovy, mackerels (+ non-CPS crab, salmon, tuna); purse seine. Segments from cluster analysis (Quezada et al., 2024, CJFAS).

Fleet segment	Main target(s)	Trips N	Vessels/yr
S. CCS industrial squid spec.	Market squid	29,160	42
Roving squid–sardine gen.	Squid & sardine	6,806	14
PNW sardine specialist	Pacific sardine	2,581	6
S. CCS forage-fish diverse	Diverse CPS	2,481	5

> Same shock, **different portfolios** — the unit of adaptation is the *fleet segment*.

The model — nested logit over *participate* × *target* × *port*

Utility of alternative $j = (p, s, l)$ for vessel i on day t:

$$U_{itj} = \delta_j + \beta_{1,s}\,\mathrm{Avail}_{tj} + \beta_{2,g}\,\mathrm{Price}_{tj} + \ldots + \varepsilon_{itj}$$

- Avail = the daily **SDM** index from the previous slide.
- Also: price, distances, wind, closures, unemployment.
- **State dependence** (last 30 days): processor ties / port capacity.
- Nests: **participation**, then **species** — except forage-diverse, which nests by **port**.
- Crab/salmon split off (costly gear switch).
- Dominant species = target (**93.6%** of revenue).

What drives the choice? Availability coefficients by segment

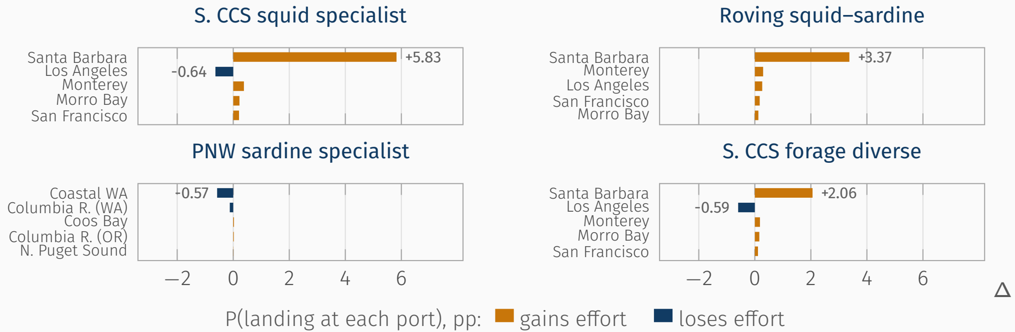
Availability of...	Squid spec.	Roving gen.	PNW sardine	Forage diverse
Market squid	+2.45***	+1.75***	—	+1.61*
Northern anchovy	+1.35***	+4.19***	+1.16***	**−1.88*****
Pacific sardine	−1.78**	0.23	−0.59**	0.23
Chub & jack mackerel	−3.94***	−1.48	0.87	−1.92***
Non-CPS (crab/salmon)	—	+5.74***	+10.3***	—
Expected price (CPS/tuna)	0.26	+4.60***	+3.21***	+0.72***
State dependence (30-day)	+2.63***	+2.86***	+2.26***	+3.09***

Wind, distance-to-catch, distance-to-CG all (−); unemployment (+). Negative anchovy for forage-diverse: anchovy–squid availability correlate $r = -0.29$. Mackerel is negative in all three southern segments (*fallback* target).

> Habit is **strong** (state dependence ≈2.3–3.1), yet the **SDM signal survives it**. Flexibility we measure is genuine, not inertia.

Move the habitat field, see where the effort goes

Same-season, squid change centroid (**42.99° N** → **40.66° N**); prices, costs, weather and state dependence fixed.



Δ **no-participation**: −6.0 / −4.2 / +0.6 / −1.9 pp · same run as Figs. 5–6 of the paper, here by port.

Santa Barbara absorbs the squid. Los Angeles loses −0.6: it *gains* squid but **loses mackerels**. Port aggregate hides a **species recomposition**.

“On the move” vs. “in place” substitution

Adapt *on the move* (species nests)

- Low $\lambda \Rightarrow$ strong correlation across **ports**, same species.
- Anchovy $\lambda = 0.49$; sardine $\lambda = 0.40$.
- $\Rightarrow$ **follow the species across ports.**

Adapt *in place* (port nests)

- Forage-diverse nests by **port**: Santa Barbara $\lambda = 0.49$ vs. LA/Monterey 0.73.
- Substitution across **species within a port**.
- $\Rightarrow$ closest substitute is **another species**, not another port.

Lower λ = alternatives inside the nest are **closer substitutes** ($\lambda = 1$: no nesting). • Terms after Samhouri et al. (2024).

> In both cases the aggregate outcome is a **spatial reallocation of effort** between ports, but the mechanism differs: one follows the species, the other substitutes species in place.

Takeaways

1. **Oceanographic SDMs work as economic regressors.** Exogenous, gap-free and daily — and the estimated models predict a **held-out year** well (OOS pseudo-ρ^2 **0.22–0.44** against the null).
2. **Harvesters are flexible.** Participation, target and landing all respond to availability, prices, costs and weather — *even* controlling for strong state dependence.
3. **Spatial redistribution can be explained by species substitution.** Even the fleet that adapts *in place* — substituting **species**, not ports — redistributes effort **in space**.

Departamento de Economía
Universidad de Concepción

INCAR^2
Interdisciplinary Center for Aquaculture Research - Applied Research

Thank you! / ¿Gracias?

Felipe J. Quezada-Escalona

felipequezada@udec.cl

felipequezada.com

Quezada-Escalona et al. (2026), *Ecological Economics* **247**

NOAA CAFA (NA20OAR4310507) · ANID-Chile FONDECYT Iniciación N° 11250223 · INCAR^2 CIA250009.

Backup • Descriptives — persistence *and* switching

	Species		Ports	
Fleet segment	Switches	Unique	Switches	Unique
S. CCS industrial squid specialist	**19.8**	2.6	**20.7**	3.3
Roving squid–sardine generalist	4.6	1.7	9.5	2.7
PNW sardine specialist	3.8	2.4	4.7	2.1
S. CCS forage-fish diverse	10.8	2.4	5.1	2.0

Vessel-level means, 2013–2017. Squid specialist: **switch rate 0.36**, **median run 2 days** — persistence *and* regular switching.

Short-run **persistence** + frequent **switching** ⇒ recent choices matter (motivates state dependence).

Backup • Elasticities — price drives *entry*, availability *targeting*

Extensive margin (participate)

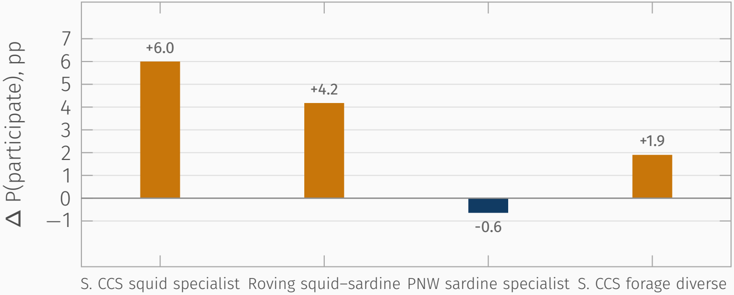
- Price dominates for mobile fleets: squid **1.71** (roving); anchovy **0.43** (PNW).
- Non-CPS availability matters where crab/salmon are relevant: crab **0.24**, sockeye **0.19** (PNW).

Intensive margin (conditional target)

- Availability often $>$ price: anchovy **2.55**, crab **2.12** (roving); crab **3.85**, sockeye **3.71** (PNW).
- Several $> 1 \Rightarrow$ more-than-proportional responses.
- Exception: forage-diverse is **price**-driven.

Prices move participation; **availability** moves which species, once at sea.

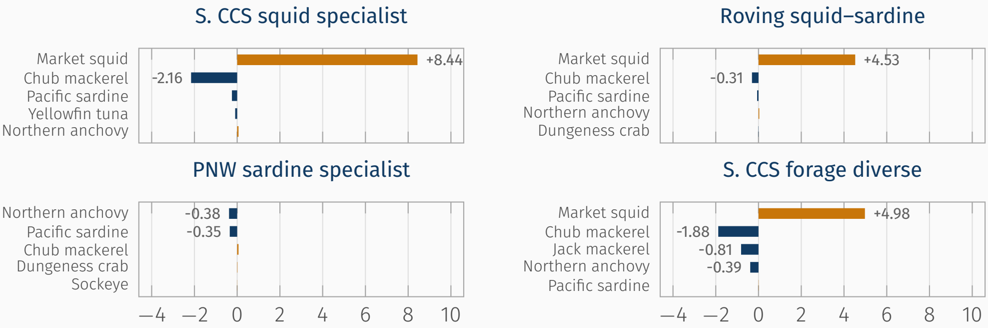
Backup • Scenario — change in *participation*



Change in the probability of **participating** (i.e. $-\Delta P$(no participation)) under the same-season window swap (southward squid redistribution). Generated from `scenario_dynamic_same_months_delta_participation_pp.csv`.

All three southern, squid-facing fleets go fishing **more often** when squid moves toward them; the PNW sardine fleet is essentially unaffected.

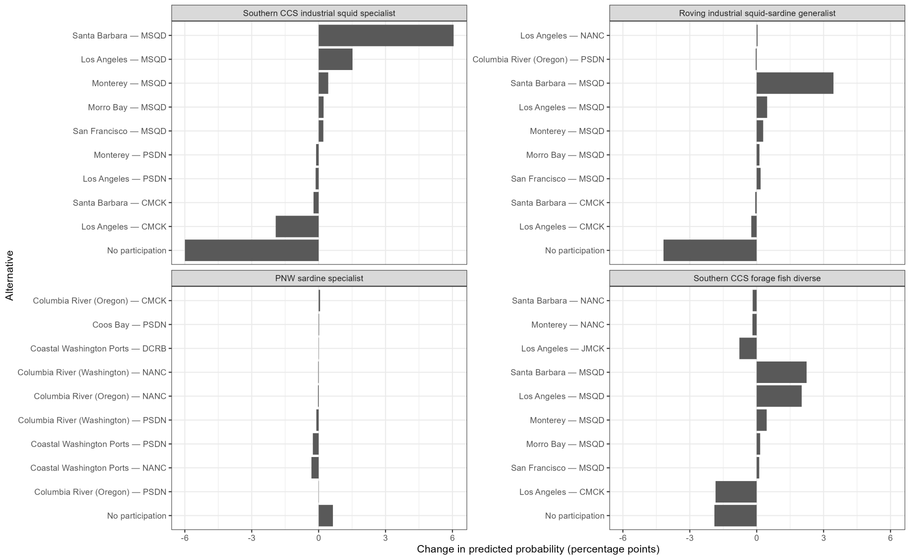
Backup • Scenario — change by *target species*



Δ predicted probability (pp): more likely targeted less likely • common axis.

Southern fleets **follow the squid** (+8.4 and +5.0 pp), and **mackerel is what they give up** (chub −2.2 and −1.9); the PNW sardine fleet barely moves — it has no squid to follow.

Backup • Scenario — change by port × species alternative



Backup • Dissimilarity parameters (λ)

Nest	Squid spec.	Roving gen.	PNW sardine	Forage diverse
Participation (CPS/Tuna)	0.977	0.752	0.532	0.730
Market squid	0.777	0.752	—	—
Pacific sardine	0.617	**0.403**	**0.404**	—
Northern anchovy	**0.494**	—	0.532	—
Santa Barbara (port)	—	—	—	**0.490**
Los Angeles / Monterey (port)	—	—	—	0.730

Lower λ = stronger within-nest correlation (more substitution within the nest).

Backup • Variable construction

- **Availability:** binomial GAM SDMs (Muhling et al., 2019, 2020); mean presence within a species radius (60 km sardine/anchovy/mackerel, 90 km squid). CPUE z-score $\rightarrow$ sigmoid for non-SDM species.
- **Price:** predicted from $\ln(\text{price}) = \alpha + T_y + \gamma_m + \gamma_p$ (year trend, month & port FE); robustness with 30-day moving average.
- **Costs:** Haversine distance port$\rightarrow$catch area and port$\rightarrow$center of gravity.
- **Weather:** max ERA5 surface wind within 220 km. **Opportunity cost:** state unemployment.

Backup • Limitations

- **Aggregation bias:** port-area resolution may mask finer spatial heterogeneity (Dépalle et al., 2021).
- **Choice-set truncation:** rare alternatives folded into the outside option ($\geq$0.5% filter); IIA assumed for omitted alternatives.
- **Price endogeneity:** predicted / MA prices reduce but do not eliminate simultaneity $\Rightarrow$ interpret price coefficients as **reduced-form** behavioral responses.
- **SDM proxy:** habitat suitability $\neq$ catch rate exactly $\Rightarrow$ availability effect may be attenuated.

Main qualitative result — **availability drives targeting** — is robust across price specifications.